AI Challenge Competition #1

Funding EU talent deeply involved with Gen AI models

Aleksandar & Daniel + 3

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Overview

About AI Challenge Competition #1

Funding

Bilal Ahmed

Update page

Questions

Admin

1 **Full Name / Team Name / Organization Name / Consortium Name ***

Bilal Ahmed

2 **Email Address ***

bk632723@gmail.com

3 **Phone Number ***

4 **Country of Residence / Registration ***

Spain

5 **Legal Status ***

Individual
SME
University
Research institution
NGO

6 **In case you are applying as a consortium, please elaborate on the consortium structure**

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EXCELLENCE

7 **Project Title ***

Sim2Field -- Multilingual Digital-Twin-Verified Mission Generation for Agricultural Robots with fa

8 **What specific problem does your AI solution aim to solve? ***

Challenge 1: GenAI-Based Natural Language Mission Generator f...

9 **Problem Fit & Contribution ***

Explain how your AI solution effectively addresses the identified problem/challenge and benefits key stakeholders, ensuring alignment with industry needs.

Agricultural robots can navigate and sense, but defining missions still requires ROS knowledge, scripting and robot-specific interfaces, excluding many farmers and small operators. Direct LLM-to-robot control is unsafe: informal or multilingual instructions can be ambiguous, miss visual or map context, or violate robot and field constraints.

Sim2Field provides a controlled mission layer: the farmer speaks or types; multimodal AI interprets the intent and grounds it in vineyard maps, visual observations, robot capabilities and current state; deterministic robotics checks verify the result; the farmer reviews and approves it; and a ROS1 adapter executes it on the Challenge Owner's platform.

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10 **Quantifiable Outcomes ***

Define KPIs, performance benchmarks, and measurable improvements expected from your solution. Provide data-driven projections demonstrating impact on efficiency, accuracy, or scalability. (KPIs, benchmarks, performance improvement, etc.).

All figures below are Advance-Phase targets, not achieved results.

KPI1 - Intent Recognition Accuracy: >=90% of predefined test instructions must have the correct task, target, location, parameters and constraints compared with ground truth.

KPI2 - Task Planning and Execution Success: >=85% of generated ROS-compatible plans must match an accepted task sequence and complete the intended mission in simulation or the controlled vineyard environment.

KPI3 - Natural-Language Robustness: >=85% success across at least 3-5 semantically

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11 **Innovation & Novelty ***

Describe the unique elements of your approach compared to existing technologies. Highlight the novelty in methods, algorithms, applications, or integrations. Specify how your solution overcomes limitations of current systems or introduces groundbreaking improvements in AI methodologies.

Sim2Field does not let an LLM turn a sentence directly into ROS commands. Its main innovation is the Mission Passport: a structured, versioned and evidence-linked mission record created before any robot action. It connects the farmer's request, extracted parameters, vineyard-map and visual grounding, robot capabilities, constraints, validation, approval and execution status.

This design separates two jobs that should not be confused. Generative AI handles flexible language and multimodal interpretation. Deterministic robotics software decides whether the proposed mission is complete, supported and safe. The LLM cannot create arbitrary code or bypass the robot's capability manifest.

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12 **Technology Readiness ***

Describe your current TRL (e.g., TRL2) and justify your ability to reach TRL5.

Sim2Field is currently at TRL3: the main technical principles have been demonstrated in separate components, but the complete speech/text-to-ROS workflow has not yet been integrated or validated on the Challenge Owner's robot.

What exists now: AgriSim is an operational agricultural robot simulator with configurable fields, robot motion, cameras, LiDAR, IMU, odometry, battery state, collision events, environmental controls and blocked or wet areas. It supports repeatable mission tests, safety events and fault injection. The applicant also has component-level work in ROS interfaces and LLM-based interpretation of robot state.

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13 **AI Technology & Justification ***

Detail the specific AI techniques, architectures, or models supporting your proposal. Explain why these technologies are ideal for your solution and how they will enhance efficiency, accuracy, and scalability. Discuss how AI interacts with complementary technologies, such as cloud computing or edge AI, to boost performance.

Sim2Field uses a modular vision-language-action pipeline so that each AI output can be checked before it affects the robot.

1.Speech and language: a multilingual Whisper-class speech model converts voice to text. A compact instruction-tuned LLM then identifies the task, target, location, parameters and constraints. Candidate 7B-class models will be benchmarked on the official tasks; the choice will be based on accuracy, latency and local-deployment feasibility rather than fixed in advance.

2.Visual and map grounding: a lightweight model detects vineyard objects from the provided imagery. Detections are associated with robot pose, available depth and named map rows or

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14 **Trustworthy, Responsible AI & Compliance ***

Summarize how your AI model ensures robustness, transparency, and explainability. Demonstrate adherence to EU AI risk classification and regulatory frameworks, including responsible AI principles such as human-in-the-loop oversight, bias mitigation, and ethical evaluation.

Sim2Field is designed so that robot safety does not depend on the LLM always being correct. The model proposes a Mission Passport; it does not publish motor commands, generate executable scripts or change safety settings. Its output is restricted to a versioned capability manifest and must pass deterministic validation before it can reach ROS.

Robustness uses schema validation, confidence thresholds, clarification for missing or contradictory information, map/visual consistency checks, simulation rehearsal and tests across language and vineyard conditions. Independent monitors supervise localization, battery, collision risk, sensors and mission state. A failed check leads to rejection, safe stop, return-to-base or human assistance under predefined rules.

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15 **Ethics & Data Governance ***

Explain how your solution manages data ethically and complies with GDPR. Provide insights into data collection, processing, security, and privacy-by-design principles. Address bias mitigation, informed consent procedures, and dataset integrity to ensure responsible AI development.

Sim2Field will maintain a data register covering source, licence, version, purpose, preprocessing and train/validation/test assignment. Challenge and public datasets will retain their original licences. The official 70/15/15 split will be applied once, with the test set held back from model selection. Duplicate checks will reduce leakage; model/data cards will document grape-type, lighting, season and viewpoint limitations.

Vineyard imagery is described as non-personal, but speech and operator logs may identify a person. Raw audio will not be retained by default. If audio or user-study data must be kept, participants will receive clear information and give informed consent. Data will be minimised, pseudonymised, access-controlled and deleted under a defined retention schedule

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IMPACT

16 **Sustainability & Scalability AI ***

What steps will ensure sustainability and continued impact of the AI model? Discuss plans for continued retraining, ethical data reuse, community engagement, and legal compliance (e.g., GDPR, AI Act). Provide a roadmap for exploitation and adaptation to evolving challenges. Include references to transparency, accountability, and data privacy in design, deployment, and stakeholder interactions.

Sim2Field is designed as a reusable mission-assurance layer rather than a one-off vineyard demo. The stable element is the Mission Passport. Robot-specific details are isolated in two replaceable modules: a capability manifest describing allowed actions and constraints, and an execution adapter connecting those actions to ROS. Supporting another robot should therefore require interface mapping and validation, not rebuilding the language pipeline.

The internal mission representation is language-independent. New spoken or written languages can be added and tested while the robot receives the same structured actions. Map entities and vision models can also be replaced for orchards, greenhouses, field inspection or industrial maintenance without changing the safety and approval logic

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17 **Social & Environmental Impact ***

What are the expected societal, environmental, and economic benefits of your solution? Connect your project to EU Green Deal policies or SDGs, demonstrating contributions to economic, social, or environmental sustainability. Include AI-driven energy efficiency strategies, emissions reduction, or infrastructure enhancements.

The primary social benefit is accessibility. A vineyard operator can define a routine mission by speaking or typing, review what the robot understood and approve it without learning ROS. Same-language clarification and the inspectable Mission Passport reduce technical barriers while keeping the operator responsible for the decision.

Economic benefit will be tested against the Challenge Owner's GUI/manual baseline using mission-definition time, correction effort and successful completion. Faster setup and fewer invalid missions could make agricultural robots more practical for SMEs and reduce specialist support for routine changes.

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18 **Fairness, Diversity & Open Science ***

How are gender, diversity, and inclusion considered in the dataset of your project? Mention inclusive team composition, stakeholder engagement, and accessibility of data or results through open science principles.

Fairness concerns who can use the interface and when vision works reliably. Gender is not a meaningful label for grape imagery and will not be inferred. Perception tests will examine available factors such as grape type, lighting, season, viewpoint and class balance. Weak conditions will trigger calibration, targeted augmentation, abstention or another observation.

Each reference mission will be tested with at least 3-5 equivalent formulations, with results separated by supported language, speech/text mode and phrasing. If ethically approved participants are available, usability testing will seek variation in gender, age, accent, language and robotics experience.

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19 **Exploitation, and Potential for Collaboration ***

What are your plans to exploit and scale the solution beyond TRL 5? Explain how the AI solution is structured for adaptability and industry-wide adoption, including plans for cross-sector collaborations, open-access research contributions, and public-private partnerships.

The first exploitation goal is a reliable integration with the Challenge Owner's vineyard robot, not a broad product claim. Sim2Field will show that a non-technical operator can create a traceable, verified mission through speech or text and execute it through the existing ROS1 stack. Evidence will include the official KPIs, comparison with GUI/manual configuration, integration effort and a representative vineyard demonstration.

After TRL5, the architecture can be offered as mission-assurance middleware for robot manufacturers and integrators. A deployment package would contain the Mission Passport schema, capability template, validators, approval interface, benchmark and ROS adapter

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Options to assess include integration services, licensing and support for validated robot profiles;

QUALITY OF IMPLEMENTATION

20 **Work Plan ***

Outline the roadmap for AI solution development, detailing key milestones and validation strategies. Explain how each phase contributes to reaching TRL5, ensuring feasibility and scalability.

Month 1 - Freeze the mission contract and baseline. With Intellimech and JOiiNT LAB, select representative vineyard missions: row navigation, grape inspection/image capture, constraint handling and return-to-base. Finalise the Mission Passport, capability manifest and action vocabulary. Confirm ROS1 and safety interfaces early. Build the instruction benchmark, 3-5 paraphrases per mission, GUI/manual baseline and KPI scoring. Connect AgriSim to mission state, map, battery, collision and logging. Milestone: hand-authored Passports execute repeatably in AgriSim and KPI scoring works.

Month 2 - Build language and grounding. Integrate speech-to-text, LLM extraction, clarification

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and visual/map grounding. Adapt the grape model using the official split. Add versioning and

21 **Do you need access to CINECA Infrastructure?**

Yes
No

22 **Resources/Infrastructure ***

Please indicate CINECA resources and infrastructure you will use.

> Partition name
> Wall clock time of a typical job execution
> Amount of GPU/CPU hours requested
> Max TB storage needed
> Software requirements
> How many models do you need to train? Parameters?
> Are there previous tests available on scalability and performance computed on similar devices as the ones available on CINECA infrastructure?

CINECA is not required for the critical path. Sim2Field will adapt existing moderate-size models rather than train a foundation model from scratch. Local development uses an NVIDIA RTX 4060 laptop GPU for 7B-class quantised LLM inference, speech recognition and lightweight perception training. Parameter-efficient tuning will be used only if structured prompting and retrieval do not meet the KPIs.

The software stack will include Linux, Docker, Python, PyTorch/Transformers, faster-whisper, OpenCV, JSON Schema/Pydantic, geospatial libraries, ROS2 for AgriSim and ROS1 tools for the Challenge Owner integration. Git-based version control, automated tests and experiment tracking will preserve reproducibility.

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23 **Team Credentials & Expertise ***

Provide evidence of the team's technical experience, achievements, and domain expertise. Highlight participation in AI research, industry collaborations, publications, competitions, and previous impactful projects.

Bilal Ahmed is applying as a solo applicant. His Erasmus Mundus training in Intelligent Field Robotic Systems (IFRoS) covers the technical areas required by Challenge 1: ROS, autonomous navigation, perception, localisation, sensor integration, robot decision-making and field deployment.

His strongest evidence of delivery is AgriSim, an operational agricultural robot simulator developed for configurable field-robot experiments. It includes robot motion, RGB and depth cameras, LiDAR, IMU, odometry, battery monitoring, collision events, environmental controls, blocked or wet areas and fault injection. This is directly useful to Sim2Field as a pre-execution verification and regression environment. It reduces early dependence on the physical platform

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24 **Risk Management & Mitigation ***

Identify key risks to implementation and how you'll address them.

1. Unsafe/unsupported AI action: constrain output to the Mission Passport and capability manifest; deterministic validation blocks invalid actions before ROS execution.
2. Speech/multilingual misunderstanding: use ASR confidence, read-back and same-language clarification; safety-critical ambiguity cannot execute.
3. Incorrect farmer-specific learning: store only approved aliases/preferences; safety rules and robot limits cannot be learned away.
4. Incorrect fault diagnosis: restrict diagnosis to defined fault classes and bounded recovery actions; uncertain or safety-critical cases trigger safe stop or human assistance.
5. Sim-to-real gap: use AgriSim-X for logic, regression and fault injection, then recalibrate perception and recovery thresholds on provided and real platform data.

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DECLARATIONS

25 **Please check the following boxes ***

- [ ] I declare that my organisation is not subject to exclusion grounds under EU Financial Regulation 2018/1046.
- [ ] If selected as a prize-winner, I agree to sign the official Declaration of Honour available on the AI-BOOST website.
- [ ] I acknowledge that selected beneficiaries may be subject to audits and controls by the European Commission, ECA, EPPO and OLAF.

Are you done? Click below to finalize

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